动态随机一般均衡模型：理论与应用
第 4 章：生产性资本

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A 动态系统

$$\beta \mathbb{E}_t \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} = \frac{1}{\hat{R}_t}$$
$$N_t^{\varphi} C_t^{\sigma} = \hat{W}_t$$
$$\mathbb{E}_t \hat{R}_{t+1}^k = \hat{R}_t + \delta$$
$$K_{t+1} = (1-\delta)K_t + I_t$$
$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}$$
$$\hat{MC}_t = \mu$$
$$\hat{W}_t = \frac{(1-\alpha)\hat{MC}_t Y_t}{N_t}$$
$$\hat{R}_t^k - 1 = \frac{\alpha \hat{MC}_t Y_t}{K_t}$$
$$Y_t = C_t + I_t$$

B 求解模型稳态

B.1 稳态系统

$$\beta = \frac{1}{\hat{R}}$$
$$N^{\varphi} C^{\sigma} = \hat{W}$$
$$\hat{R}^k = \hat{R} + \delta$$
$$(\hat{R}^k - 1)K = \alpha \hat{MC} Y$$
$$Y = AK^{\alpha} N^{1-\alpha}$$
$$\hat{MC} = \mu$$
$$\hat{W} N = (1-\alpha)\hat{MC} Y$$
$$Y = C + I$$
$$K = (1-\delta)K + I$$

B.2 稳态均衡

$$\hat{R} = \frac{1}{\beta}$$
$$\hat{R}^k = \hat{R} + \delta$$
$$\hat{MC} = \mu$$
$$\hat{W} = (1-\alpha)\mu\Theta^{\frac{\alpha}{1-\alpha}}$$
$$Y = \left[(1-\alpha)\mu\Theta^{\frac{\alpha(1+\varphi)}{1-\alpha}}(1-\delta\Theta)^{-\sigma}\right]^{\frac{1}{\sigma+\varphi}}$$
$$N = \Theta^{-\frac{\alpha}{1-\alpha}}Y$$
$$K = \Theta Y$$
$$I = \delta\Theta Y$$
$$C = (1-\delta\Theta)\,Y$$

其中，$\Theta = \frac{\alpha\mu}{\hat{R}^k-1}$。

C 对数线性化

$$\sigma(\mathbb{E}_t c_{t+1} - c_t) = \hat{r}_t$$
$$\varphi n_t + \sigma c_t = \hat{w}_t$$
$$\hat{R}^k \mathbb{E}_t \hat{r}^k_{t+1} = \hat{R}\hat{r}_t$$
$$k_{t+1} = (1-\delta)k_t + \delta i_t$$
$$y_t = a_t + \alpha k_t + (1-\alpha)n_t$$
$$\hat{w}_t + n_t = y_t$$
$$\hat{R}^k \hat{r}^k_t = \frac{\alpha\mu}{\Theta}(y_t - k_t)$$
$$y_t = (1-\delta\Theta)c_t + \delta\Theta i_t$$